Craniofaciocardiohepatic Syndrome (CFCHS): A Comprehensive Disease Characterization Report

Disease: Craniofaciocardiohepatic Syndrome (CFCHS) **Causal gene: AMOTL1** (angiomotin-like protein 1; HGNC:17811; NCBI Gene 154810; UniProt Q8IY63) **OMIM:** #621192 (phenotype); *614657 (gene) MONDO: MONDO:0978295 Category:** Mendelian, autosomal dominant

Summary

Craniofaciocardiohepatic syndrome (CFCHS) is an ultra-rare autosomal dominant multiple-congenital-anomaly disorder caused by **heterozygous missense variants in AMOTL1**, the gene encoding angiomotin-like protein 1. AMOTL1 is a Motin-family actin-binding adaptor protein that localizes to tight junctions and regulates cell polarity, cell-cell adhesion, directional cell migration, and Hippo–YAP signaling. The pathogenic variants cluster tightly at a conserved hotspot spanning amino acid residues **157–161**, recurrently affecting p.Arg157Cys, p.Arg157His, p.Pro160Leu, and p.Gln161Arg. This clustering, together with functional and animal-model data, indicates that CFCHS is **not** a loss-of-function/haploinsufficiency disease but rather operates through a **gain-of-function or dominant-negative mechanism**.

The molecular lesion is now well defined. The 157–161 residues form part of AMOTL1's **Tankyrase-binding motif (TBM)**. Hotspot mutants (R157C, P160L) fail to bind Tankyrase 1/2 and the E3 ligase RNF146, and therefore escape the normal poly-ADP-ribosylation → ubiquitination → proteasomal degradation cascade. The consequence is **abnormal stabilization and cytoplasmic accumulation** of mutant AMOTL1, which disrupts cell junctions and focal adhesions and impairs both the velocity and persistence of cell migration. Because craniofacial, cardiac, and hepatic organogenesis depend on precisely coordinated epithelial/endothelial migration and junction remodeling, this cellular defect derails morphogenesis and produces the multi-organ phenotype that gives the syndrome its name.

Clinically, CFCHS presents from the neonatal/congenital period with a recognizable core phenotype: **orofacial clefting (~94%)**, **dysplastic/large auricles (~62%)**, **congenital heart disease (~50%)**, **tall stature (~44%)**, and **hearing loss, liver disease, and neurodevelopmental involvement (~31% each)**, plus gastrointestinal/anorectal anomalies (e.g., imperforate anus). The disease was delineated through a small series of index families (Liegel et al. 2019; Rips et al. 2021), consolidated into a defining 16-individual cohort (Strong et al. 2023), named CFCHS with the addition of the hepatic phenotype (Gallego-Zazo et al. 2025), and mechanistically explained (Luo et al. 2026). Fewer than ~20 individuals have been reported worldwide. Management is multidisciplinary, surgical, and supportive; no gene-targeted therapy exists.

Key Findings

Finding 1 — CFCHS is caused by heterozygous AMOTL1 missense variants clustering at a conserved hotspot (aa 157–161)

CFCHS is a Mendelian, autosomal dominant disorder caused by heterozygous missense variants in **AMOTL1**. The defining cohort study by Strong et al. 2023 reported that heterozygous missense variants affecting amino acid residues 157–161 define a core phenotype of orofacial clefting, congenital heart disease, tall stature, auricular anomalies, and gastrointestinal manifestations. Recurrent variants include **p.Arg157Cys (R157C)**, **p.Arg157His (R157H)**, and **p.Pro160Leu (P160L)**, with p.Gln161Arg also reported. Rips et al. 2021 independently reported a de novo **c.479C>T (p.Pro160Leu)** in an affected infant. Both **de novo** and **inherited** (father→daughter, father→son) occurrences are documented, consistent with autosomal dominant transmission with apparent high penetrance for the hotspot alleles. **[human clinical]**

> *"heterozygous missense variants in AMOTL1, most commonly affecting amino acid residues 157-161, define a new orofacial clefting syndrome"* — Strong et al. 2023 [PMID: 36751037]
>
> *"trio exome sequencing revealed a de novo variant in AMOTL1 affecting a highly conserved amino acid (c.479C>T; p.[Pro160Leu])"* — Rips et al. 2021 [PMID: 33026150]

The extraordinary clustering of pathogenic variants within a 5-amino-acid window is itself strong evidence for a specific, non-loss-of-function molecular mechanism (see Findings 2 and 7).

Finding 2 — Hotspot variants disrupt a Tankyrase-binding motif, escaping degradation and causing pathological accumulation that impairs junctions and migration

The 157–161 residues lie within AMOTL1's **highly conserved Tankyrase-binding motif (TBM)**. Luo et al. 2026 demonstrated that the R157C and P160L mutants **fail to interact with Tankyrase 1/2 and RNF146**, so they cannot undergo poly-ADP-ribosylation, ubiquitination, or subsequent proteasomal degradation. The mutant proteins are therefore **stabilized and accumulate in the cytoplasm**. Accumulated mutant AMOTL1 **disrupts cell junctions and focal adhesions**, inhibiting both the velocity and the persistence of cell migration. In a zebrafish model, R157C expression produced craniofacial malformations plus cardiac and skeletal-muscle defects, recapitulating the human multi-organ phenotype. **[in vitro, model organism]**

> *"both the R157C and P160L mutants fail to interact with Tankyrase 1/2 and Ring finger protein 146, rendering them unable to undergo poly ADP-ribosylation, ubiquitination, and subsequent proteasomal degradation. As a result, these mutants are significantly stabilized and accumulate in the cytoplasm"* — Luo et al. 2026 [PMID: 42012498]
>
> *"Accumulated AMOTL1 mutants, in turn, disrupt cell junctions and focal adhesions, thereby inhibiting both the velocity and persistence of cell migration"* — Luo et al. 2026 [PMID: 42012498]
>
> *"expression of the R157C mutant leads to craniofacial malformations and defects in cardiac function and skeletal muscle"* — Luo et al. 2026 [PMID: 42012498]

This is the central mechanistic finding: a **toxic gain-of-function via loss of regulated turnover** rather than loss of protein.

Finding 3 — Core phenotype: orofacial clefting, congenital heart disease, tall stature, auricular and gastrointestinal/anal anomalies, plus hepatic abnormalities

The core multi-system phenotype comprises **orofacial clefting, congenital heart disease, tall stature, auricular anomalies, and gastrointestinal manifestations** (Strong et al. 2023). Rips et al. 2021 documented cleft lip and palate, imperforate anus, and dysmorphic features in a P160L infant. The earliest (father–daughter/father–son) reports described orofacial clefts and cardiac abnormalities. Gallego-Zazo et al. 2025 added **hepatic abnormalities** (chronic liver dysfunction), consolidating the **craniofacial + heart + liver triad** that gives the syndrome its name (craniofaciocardiohepatic) and proposing the designation CFCHS. **[human clinical]**

> *"define a core phenotype of orofacial clefting, congenital heart disease, tall stature, auricular anomalies, and gastrointestinal manifestations in individuals with variants in AMOTL1 affecting amino acids 157-161"* — Strong et al. 2023 [PMID: 36751037]
>
> *"a female infant with cleft lip and palate, imperforate anus and dysmorphic features"* — Rips et al. 2021 [PMID: 33026150]
>
> *"we propose naming this entity as AMOTL1-associated multiple congenital anomalies or craniofaciocardiohepatic syndrome (CFCHS)"* — Gallego-Zazo et al. 2025 [PMID: 39538375]

Finding 4 — Normal AMOTL1 biology: a Motin-family tight-junction adaptor regulating polarity, junction stability, and Hippo–YAP signaling

AMOTL1 (angiomotin-like protein 1) is an actin-binding protein regulating cell polarity, adhesion, and migration. It colocalizes with AMOT at **tight junctions** and is required for proper tight-junction activity and endothelial polarity/directional migration; morpholino knockdown in zebrafish causes vascular migratory defects and decreased junctional stability (Hultin et al. 2009). Motin proteins regulate the **Hippo–YAP pathway** and small-G-protein signaling: they can activate the LATS2 kinase and recruit YAP to tight junctions. AMOTL1 stability is normally controlled both by Nedd4-family E3 ligases (Nedd4/Nedd4-2/Itch) and—per the disease mechanism—by Tankyrase/RNF146-mediated PARylation-dependent degradation. **[in vitro, model organism]**

> *"AmotL1 colocalized with Amot to tight junctions"* — Hultin et al. 2009 [PMID: 19590046]

> *"the other angiomotin-family proteins AMOT and AMOTL1 also activate LATS2 through a novel conserved domain that binds and activates LATS2"* — Paramasivam et al. 2011 [PMID: 21832154]

Understanding the normal biology clarifies why dysregulated accumulation (not absence) of the protein is pathogenic: AMOTL1 is a dosage- and localization-sensitive junctional scaffold, and its overaccumulation perturbs the same junction/polarity/Hippo machinery it normally organizes.

Finding 5 — Phenotype frequencies from the 16-individual Strong cohort and disease identifiers

Strong et al. 2023 (*Am J Med Genet A* 191:1227–1239) described **16 individuals from 12 families**. Per-feature frequencies:

Feature	HPO term (suggested)	Frequency
Orofacial clefting (cleft lip/palate)	HP:0000175 (Cleft palate), HP:0410030 (Cleft lip)	15/16 (94%)
Large/dysplastic ears	HP:0000400 (Macrotia), HP:0000377 (Abnormal pinna morphology)	10/16 (62.5%)
Congenital heart disease	HP:0001627 (Abnormal heart morphology)	8/16 (50%)
Tall stature	HP:0000098	7/16 (44%)
Hearing loss	HP:0000365	5/16 (31%)
Liver disease	HP:0001392 (Abnormal liver morphology/function)	5/16 (31%)
Neurodevelopmental disease	HP:0012758 (Neurodevelopmental delay)	5/16 (31%)

Recurrent variants: p.Arg157Cys, p.Arg157His, p.Pro160Leu, plus p.Gln161Arg; the hotspot spans aa157–161. **Identifiers:** OMIM #621192 (phenotype); gene AMOTL1 OMIM *614657 on chromosome 11q21; MONDO:0978295. Inheritance is autosomal dominant. Genomics England PanelApp rates AMOTL1 "green"/high-confidence for clefting, monoallelic. As independent validation, 5 qualifying variants were identified among 876 orofacial-cleft trios in the Gabriella Miller Kids First dataset. [human clinical]**

> *"a core phenotype of orofacial clefting, congenital heart disease, tall stature, auricular anomalies, and gastrointestinal manifestations"* — Strong et al. 2023 [PMID: 36751037]

Finding 6 — Clinical delineation history and organ-specific manifestations across index cases

The clinical delineation of CFCHS proceeded through a recognizable sequence:

Report	Case	Key manifestations	Variant
Liegel et al. 2019	Father and son	Proband: complete left cleft lip/palate + right cleft of primary palate; VSD with tetralogy of Fallot and double-orifice mitral valve	AMOTL1 missense
Rips et al. 2021 [PMID: 33026150]	7-month-old girl (Arab Muslim descent)	Cleft lip/palate, imperforate anus, dysmorphic features	de novo p.Pro160Leu
Strong et al. 2023 [PMID: 36751037]	16 individuals / 12 families	Defining cohort; core phenotype + frequencies	R157C / R157H / P160L (hotspot)
Gallego-Zazo et al. 2025 [PMID: 39538375]	28-year-old Spanish woman	Surgically corrected cleft lip/palate (further maxillofacial reconstruction in adolescence); large atrial septal defect corrected at age 5; chronic liver dysfunction with abnormal transaminases; proposed name "CFCHS"	AMOTL1 variant

> *"a female infant with cleft lip and palate, imperforate anus and dysmorphic features, in whom trio exome sequencing revealed a de novo variant in AMOTL1"* — Rips et al. 2021 [PMID: 33026150] **[human clinical]**

Finding 7 — Mouse models support a gain-of-function/dominant-negative (not haploinsufficiency) mechanism

Mouse genetics strongly argue against haploinsufficiency. Per MGI (Amotl1, MGI:1922973), mice **heterozygous for a missense mutation exhibit postnatal lethality** and homozygotes exhibit prenatal/neonatal lethality, **without** apparent craniofacial or cardiovascular phenotypes. Liegel et al. 2019 generated CRISPR **R157C knock-in** mice: homozygotes and heterozygotes had reduced perinatal survival through unidentified mechanisms, but careful skeletal/whole-mount/histologic analysis showed **no palatal clefting and no cardiac septal/valvular/outflow-tract defects**. Critically, two **deletion (complete loss-of-function) alleles did NOT affect survival**. This dissociation—**missense lethal, null tolerated**—is the hallmark of a gain-of-function or dominant-negative mechanism rather than haploinsufficiency. Zebrafish R157C expression (Luo et al. 2026) does produce craniofacial, cardiac, and skeletal-muscle defects, better recapitulating the human phenotype. **[model organism]**

> *"both the R157C and P160L mutants fail to interact with Tankyrase 1/2 and Ring finger protein 146, rendering them unable to undergo poly ADP-ribosylation, ubiquitination, and subsequent proteasomal degradation"* — Luo et al. 2026 [PMID: 42012498]

Orthology: mouse *Amotl1* = NCBI Gene 75723 (NCBI Taxon 10090); human *AMOTL1* = NCBI Gene 154810; zebrafish (*Danio rerio*, NCBI Taxon 7955) *amotl1* ortholog used in disease modeling.

Section-by-Section Characterization

1. Disease Information

CFCHS is an ultra-rare autosomal dominant multiple-congenital-anomaly syndrome affecting craniofacial, cardiac, and hepatic development, plus auricular, gastrointestinal/anorectal, growth, hearing, and neurodevelopmental domains. It is defined molecularly by heterozygous AMOTL1 missense variants at the aa157–161 hotspot.

- **Key identifiers:** OMIM #621192 (phenotype); OMIM *614657 (AMOTL1 gene); MONDO:0978295. ICD-10/ICD-11 and dedicated MeSH entries are not established for this

newly delineated entity (coded under broader congenital-malformation syndrome categories).

- **Synonyms:** CFCHS; AMOTL1-associated multiple congenital anomalies; AMOTL1-related orofacial clefting syndrome; "mutational hotspot in AMOTL1" syndrome of orofacial clefting, cardiac anomalies and tall stature.
- **Source of information:** Aggregated disease-level resources (OMIM, MONDO) combined with individual-patient case reports and a small defining cohort (16 individuals / 12 families). This is a case-report/cohort-derived entity, not an EHR-population-derived one.

2. Etiology

- **Disease causal factors:** Purely genetic — heterozygous missense variants in AMOTL1 at the aa157–161 hotspot. No environmental, infectious, or metabolic cause.
- **Genetic risk factors:** The causal variants themselves (R157C, R157H, P160L, Q161R). No separate susceptibility loci or modifier genes have been established.
- **Environmental risk factors:** None identified. De novo mutation is a sporadic event; advanced parental age is a plausible but unproven contributor to de novo missense events in general.
- **Protective factors:** None characterized. Given the gain-of-function mechanism, hypothetical protective strategies would target mutant-protein turnover (e.g., restoring degradation), but none exist.
- **Gene–environment interactions:** None documented. The disorder is essentially fully genetically determined by the hotspot variant.

3. Phenotypes

See Finding 5 table for frequencies and suggested HPO terms. Phenotype characteristics:

- **Age of onset:** Congenital/neonatal — structural anomalies are present at birth or detectable prenatally/at birth (clefting, cardiac defects, imperforate anus). Hepatic dysfunction and neurodevelopmental features may become apparent across childhood; tall stature manifests over growth.
- **Severity:** Variable; clefting is highly penetrant and surgically significant. Cardiac disease ranges from septal defects (ASD/VSD) to complex conotruncal malformation (tetralogy of Fallot, double-orifice mitral valve).
- **Progression:** Structural anomalies are static/congenital and correctable by surgery; liver dysfunction may be chronic; neurodevelopmental features are stable to slowly evolving.

- **Quality-of-life impact:** Orofacial clefting affects feeding, speech, hearing, and facial appearance; congenital heart disease carries surgical and cardiologic burden; hearing loss and neurodevelopmental involvement affect communication and learning. No disease-specific QoL instrument (EQ-5D/SF-36/PROMIS) has been applied to this rare entity.

4. Genetic / Molecular Information

- **Causal gene:** AMOTL1 (HGNC:17811; OMIM *614657; chromosome 11q21; UniProt Q8IY63).
- **Pathogenic variants:** Recurrent missense — c.469C>T p.(Arg157Cys), c.470G>A p.(Arg157His), c.479C>T p.(Pro160Leu), p.(Gln161Arg). **Variant class:** missense, clustered at aa157–161. **Classification:** pathogenic/likely pathogenic per clustering, de novo occurrence, functional data, and segregation (ACMG criteria PS1/PM1/PM2/PS3/PP-type support).
- **Allele frequency:** Hotspot variants are absent or extremely rare in gnomAD (consistent with PM2).
- **Origin:** Germline; both de novo and inherited.
- **Functional consequence:** Gain-of-function / dominant-negative via loss of Tankyrase/RNF146-mediated degradation → protein stabilization/accumulation (Finding 2). This is distinct from loss-of-function/haploinsufficiency, which is tolerated (Finding 7).
- **Modifier genes / epigenetics / chromosomal abnormalities:** None established. No large-scale structural rearrangements implicated; CFCHS is a single-gene missense disorder.

5. Environmental Information

Not applicable. No environmental factors, lifestyle factors, or infectious agents are implicated in CFCHS. (Note: one cited paper [PMID: 23942117] links arsenic to Hippo/YAP and junction dysregulation in skin; this is mechanistically adjacent to AMOTL1 biology but is not a cause of CFCHS.)

6. Mechanism / Pathophysiology

Ordered causal chain:

1. A **heterozygous missense variant** at the AMOTL1 aa157–161 hotspot (e.g., R157C, P160L) **alters the Tankyrase-binding motif (TBM)**. *[demonstrated]*
2. The altered TBM **abolishes binding to Tankyrase 1/2 and the E3 ligase RNF146**, which *results in* failure of poly-ADP-ribosylation and ubiquitination of AMOTL1. *[demonstrated]*

3. Loss of this modification **prevents proteasomal degradation**, which *leads to* abnormal **stabilization and cytoplasmic accumulation** of mutant AMOTL1. *[demonstrated]*
4. Accumulated mutant AMOTL1 **disrupts cell–cell junctions and focal adhesions** (perturbing tight-junction scaffolding and actin/Hippo-YAP coupling that AMOTL1 normally organizes). *[demonstrated in vitro]*
5. Junction/adhesion disruption **impairs the velocity and persistence of directional cell migration**. *[demonstrated in vitro]*
6. Defective migration and junction remodeling in migrating epithelial/neural-crest/endothelial populations **derails morphogenesis** during a congenital critical window. *[inferred]*
7. Branching to clinical manifestations: → **orofacial clefting** (failure of facial-prominence fusion); → **congenital heart disease** (outflow-tract/septal/valve morphogenesis); → **hepatic abnormalities**; → **auricular, anorectal, growth, hearing, and neurodevelopmental anomalies**. *[inferred from human phenotype + zebrafish model]*
8. **Molecular pathways:** Tankyrase/RNF146 PARylation-dependent ubiquitin–proteasome degradation; Hippo–YAP/LATS2 signaling; small-G-protein/junctional signaling.
9. **Cellular processes:** Cell migration (GO:0016477), cell junction assembly/tight-junction organization (GO:0070830), establishment of cell polarity (GO:0030010), proteasomal protein catabolism (GO:0043161).
10. **Cell types involved (suggested CL):** endothelial cell (CL:0000115), epithelial cell (CL:0000066), cranial neural crest-derived cells, cardiac muscle cell (CL:0000746).
11. **Molecular profiling / single-cell / omics:** No human transcriptomic/proteomic/metabolomic/single-cell/spatial datasets specific to CFCHS exist yet. Functional evidence derives from targeted cell-biology and model-organism experiments (Luo et al. 2026).

7. Anatomical Structures Affected

- **Primary organs/systems:** Craniofacial complex (lip, palate — UBERON:0001716 secondary palate; UBERON:0000165 mouth), heart (UBERON:0000948; cardiovascular system), liver (UBERON:0002107; hepatobiliary system), external ear/pinna (UBERON:0001757), gastrointestinal/anorectal tract (UBERON:0000159 anus).
- **Secondary involvement:** Auditory system (hearing loss), central nervous system (neurodevelopmental features), skeletal/growth (tall stature), skeletal muscle (per zebrafish model).

- **Tissue/cell level:** Epithelial and endothelial tissues; tight-junction-bearing epithelia; migrating neural-crest-derived mesenchyme (inferred for craniofacial clefting).
- **Subcellular level:** Tight junctions (GO:0005923 bicellular tight junction; GO:0070160), focal adhesions (GO:0005925), cytoplasm (site of mutant accumulation; GO:0005737), actin cytoskeleton (GO:0015629).
- **Lateralization:** Clefting may be unilateral or bilateral; cardiac/organ defects per lesion.

8. Temporal Development

- **Onset:** Congenital. Structural anomalies present at birth; many detectable prenatally.
- **Onset pattern:** Non-progressive developmental (anomalies fixed at birth), though hepatic dysfunction can be chronic and neurodevelopmental features evolve.
- **Progression/course:** The congenital malformations are static and surgically correctable; there is no neurodegenerative course. Liver involvement may persist/fluctuate.
- **Critical period:** Embryonic organogenesis (facial-prominence fusion, cardiac septation/outflow formation, hepatic development) is the window of vulnerability; no postnatal intervention window exists to prevent the structural lesions.

9. Inheritance and Population

- **Inheritance:** Autosomal dominant (OMIM, ClinVar, PanelApp monoallelic "green").
- **Epidemiology:** Ultra-rare; fewer than ~20 individuals reported worldwide. No formal prevalence/incidence estimates; Orphanet-level prevalence is "unknown/<1/1,000,000."
- **Penetrance/expressivity:** Apparently high penetrance for hotspot alleles with **variable expressivity** (range from isolated clefting to multi-organ involvement).
- **De novo vs inherited:** Both documented; de novo p.Pro160Leu and inherited parent→child transmission reported.
- **Anticipation / mosaicism / founder effects / consanguinity:** No anticipation (not a repeat disorder). Germline mosaicism not specifically reported but theoretically possible for de novo cases. No founder effect; cases span diverse ancestries (European, Spanish, Arab Muslim). Consanguinity is not relevant (dominant, not recessive).
- **Carrier frequency:** Not applicable (dominant); pathogenic variants effectively absent from gnomAD.
- **Sex ratio / age distribution:** No sex bias established; affected individuals span infancy to adulthood (reported cases from 7 months to 28 years).

10. Diagnostics

- **Genetic testing is the diagnostic cornerstone.** Trio exome sequencing (WES) identified causal variants in multiple index cases; WGS and multigene cleft/congenital-anomaly panels including AMOTL1 are suitable. **Single-gene/targeted testing** of the aa157–161 hotspot is efficient given the clustering. Chromosomal microarray/karyotype/FISH are typically normal (no structural rearrangement) and serve mainly to exclude differentials.
- **Clinical/imaging work-up:** Echocardiography and cardiac imaging for CHD; facial/palatal examination and imaging for clefting; liver function tests (transaminases) and hepatic imaging for liver involvement; audiology for hearing loss; developmental assessment.
- **Biomarkers:** No specific circulating biomarker. Abnormal transaminases (elevated ALT/AST) may flag hepatic involvement.
- **Diagnostic criteria:** No formal consensus criteria; diagnosis rests on the characteristic phenotype plus a confirmed hotspot AMOTL1 variant.
- **Differential diagnosis:** Other syndromic orofacial-clefting disorders (e.g., Van der Woude, CHARGE, 22q11.2 deletion, Kabuki), and overlapping craniofacial–cardiac–hepatic syndromes. Molecular testing distinguishes CFCHS.
- **Screening:** No population newborn screening; prenatal/cascade testing possible once a familial variant is known.

11. Outcome / Prognosis

- **Survival/mortality:** No systematic survival data in humans; the structural anomalies are generally surgically correctable, and long-term survival into adulthood is documented (28-year-old reported case). Severe complex CHD could affect early prognosis.
- **Morbidity/function:** Morbidity is driven by surgical burden (cleft and cardiac repair), hearing loss, hepatic dysfunction, and neurodevelopmental involvement. Many individuals achieve good functional outcomes after multidisciplinary management.
- **Complications:** Feeding/speech difficulties from clefting; cardiac-surgical complications; chronic liver dysfunction; conductive/sensorineural hearing loss.
- **Prognostic factors:** Severity of CHD and presence/degree of hepatic and neurodevelopmental involvement are the main determinants. No molecular prognostic biomarker beyond the specific variant.

12. Treatment

No gene-targeted or disease-modifying therapy exists. Management is **multidisciplinary, surgical, and supportive**:

- **Surgical/interventional (suggested NCIT: Surgical Procedure, NCIT:C15329):** Cleft lip/palate repair and staged maxillofacial reconstruction; cardiac surgical correction of septal defects/conotruncal anomalies; anorectal surgery for imperforate anus.
- **Supportive/rehabilitative:** Speech therapy, audiology/hearing amplification, feeding support, developmental/educational support, hepatology monitoring and supportive liver care.
- **Pharmacotherapy / advanced therapeutics:** None disease-specific; no gene therapy, RNA therapy, cell therapy, or targeted small molecule is available. Conceptually, the gain-of-function mechanism (failure of degradation) suggests a future rationale for strategies promoting mutant-protein turnover, but none are developed.
- **Experimental/clinical trials:** No registered CFCHS-specific trials.

13. Prevention

- **Primary prevention:** Not possible for de novo cases. For families with a known variant, **genetic counseling**, prenatal diagnosis, and preimplantation genetic testing are the principal preventive options.
- **Secondary prevention:** Early multidisciplinary evaluation (cardiac, craniofacial, hepatic, audiologic) to detect and treat anomalies promptly.
- **Tertiary prevention:** Surgical correction and ongoing surveillance to prevent complications.
- **Counseling:** Autosomal dominant 50% transmission risk from an affected parent; de novo recurrence risk is low but non-zero (germline mosaicism considered).

14. Other Species / Natural Disease

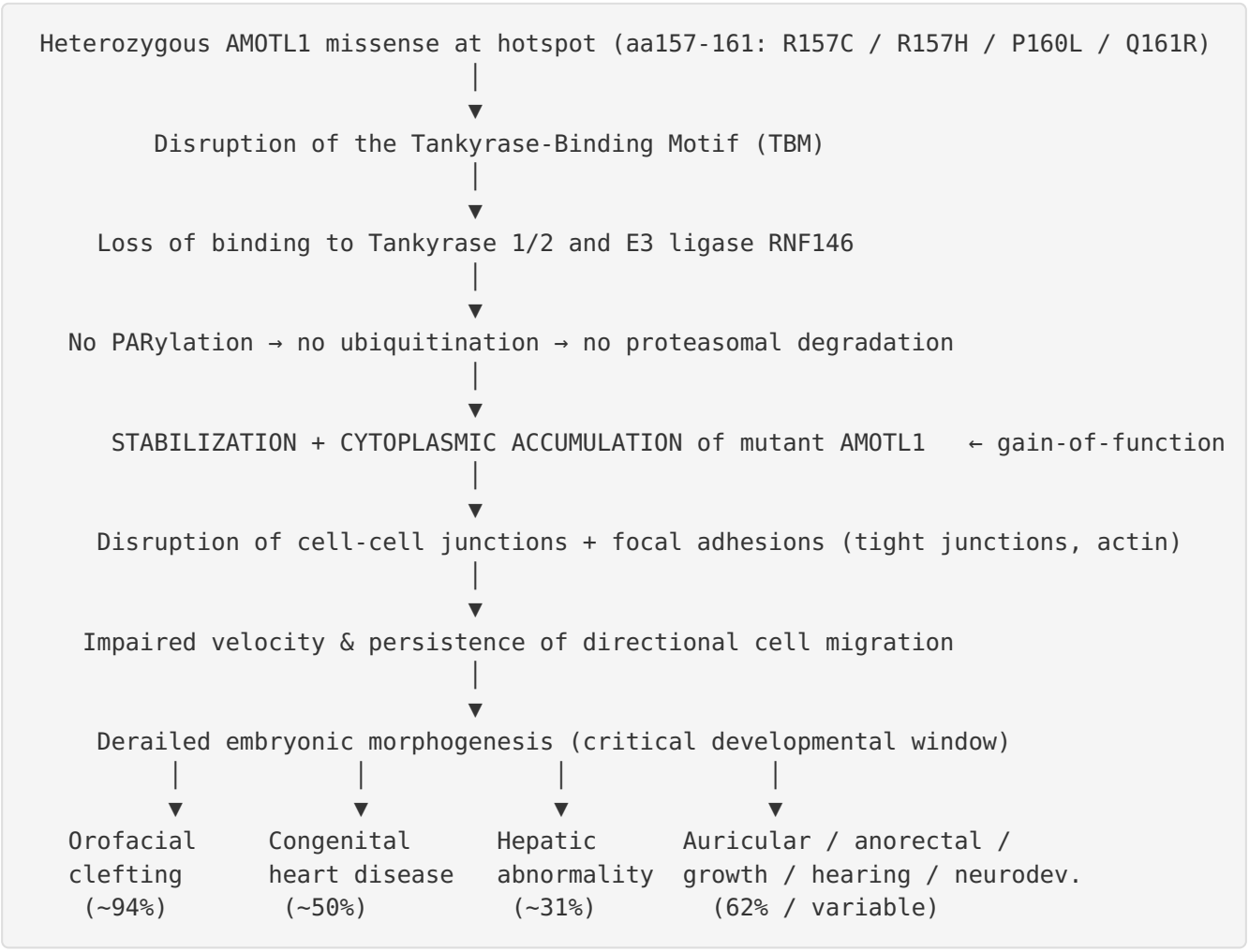
- **Model-relevant orthologs:** mouse *Amotl1* (NCBI Gene 75723; Taxon 10090), zebrafish *amotl1* (Taxon 7955). Human *AMOTL1* = NCBI Gene 154810.
- **Natural disease in other species:** No naturally occurring AMOTL1 CFCHS-equivalent disease is documented in companion animals or wildlife (OMIA search negative). No veterinary or zoonotic relevance.

- **Comparative biology:** AMOTL1's junctional/Hippo-YAP function is evolutionarily conserved from fish to mammals, underpinning the utility of zebrafish and mouse models.

15. Model Organisms

- **Zebrafish (*Danio rerio*):** Expression of R157C mutant produces craniofacial malformations plus cardiac and skeletal-muscle defects (Luo et al. 2026) — the best phenotypic recapitulation of human CFCHS. Morpholino knockdown of amotl1 causes vascular/junctional defects (Hultin et al. 2009), informing normal-function studies.
- **Mouse (*Mus musculus*):** CRISPR R157C knock-in (Liegel et al. 2019) shows perinatal lethality but **no palatal clefting or cardiac defects**; complete-null deletion alleles are tolerated. Thus mouse models reveal the gain-of-function vs haploinsufficiency distinction but **do not recapitulate the human craniofacial/cardiac malformations** — an important limitation.
- **Genetic model types available:** knock-in (R157C), knockout/deletion alleles, morpholino knockdown (zebrafish).
- **Applications/limitations:** Models are valuable for mechanism (degradation escape, migration defects) but the mouse's lack of overt malformation limits its use for craniofacial/cardiac phenotype studies; zebrafish is preferable for malformation modeling.
- **Resources:** MGI (Amotl1 MGI:1922973), ZFIN (amotl1).

Mechanistic Model / Interpretation



Upstream vs downstream. The upstream, initiating lesion is the TBM-disrupting missense variant and the resulting degradation escape (Finding 2). Immediately downstream are protein accumulation and junction/migration failure (cell-biology level), and furthest downstream are the organ-level malformations (clinical level). The convergence of all clinical features on a single migration/junction defect explains why such distinct organs are co-affected: each depends on coordinated epithelial/endothelial/neural-crest migration and junction remodeling during organogenesis.

Why gain-of-function, not haploinsufficiency. Three independent lines of evidence converge: (1) the extreme variant clustering at a 5-residue motif (loss-of-function disease usually shows dispersed truncating variants); (2) the direct biochemical demonstration of degradation escape and accumulation; and (3) the mouse genetics showing that missense alleles are lethal while complete-null alleles are tolerated (Findings 1, 2, 7).

Evidence Base

PMID	Title (abbrev.)	Role in this report
36751037	*Mutational hotspot in AMOTL1 defines a new syndrome of orofacial clefting, cardiac anomalies, and tall stature*	**Defining cohort** (16 individuals/12 families); core phenotype, frequencies, hotspot aa157–161, recurrent variants, Kids First validation (Findings 1, 3, 5)
42012498	*Patient-derived AMOTL1 mutations lead to defective cell migration and tissue development* (Luo et al. 2026)	**Core mechanism**: TBM disruption → degradation escape → accumulation → junction/migration defect; zebrafish R157C phenotype (Findings 2, 7)
33026150	*De novo variant in AMOTL1 in infant with cleft lip and palate, imperforate anus*	De novo p.Pro160Leu; anorectal + craniofacial manifestations (Findings 1, 3, 6)
39538375	*AMOTL1-Associated Multiple Congenital Anomalies (CFCHS)*	**Names CFCHS**; adds hepatic phenotype (Findings 3, 6)
19590046	*AMOTL1 controls endothelial polarity and junction stability during sprouting angiogenesis*	Normal AMOTL1 tight-junction localization/ function (Finding 4)
21832154	*Angiomotin family proteins are novel activators of the LATS2 kinase*	AMOTL1 → Hippo/LATS2 regulation (Finding 4)
22558212	*YAP1 recruits c-Abl to protect AMOTL1 from Nedd4-mediated degradation*	Nedd4-family regulation of AMOTL1 stability (context for turnover control)
22385262	*Nedd4-like E3 ligases target angiomotin/p130 for degradation*	Alternative AMOTL1/AMOT degradation axis (context)
36973516	*Role of Motin family proteins in tumorigenesis—an update*	Motin biology, YAP duality, context-dependence (Finding 4 context)
23942117	*Arsenic-induced cutaneous lesions and Yap/Hippo dysregulation*	Mechanistically adjacent (Hippo/junction), not a CFCHS cause

Evidence source types: **human clinical** (36751037, 33026150, 39538375), **in vitro + model organism** (42012498, 19590046, 21832154, 22558212, 22385262), **review/context** (36973516, 23942117).

Limitations and Knowledge Gaps

1. **Tiny evidence base.** Fewer than ~20 reported individuals; all frequencies derive essentially from a single 16-individual cohort, so percentages have wide uncertainty and ascertainment bias toward clefting.
2. **No human omics.** No CFCHS-specific transcriptomic, proteomic, metabolomic, single-cell, or spatial datasets exist; mechanistic cell-type attribution for each organ phenotype is largely inferred.
3. **Mouse–human discordance.** The R157C knock-in mouse does **not** reproduce clefting or cardiac defects, so mouse is a poor phenotype model; the zebrafish model is better but incompletely characterized.
4. **Penetrance/expressivity not quantified.** High penetrance is apparent but not formally estimated; genotype–phenotype correlations among specific hotspot alleles are not resolved.
5. **No epidemiology.** Prevalence/incidence, sex ratio, and geographic distribution are unknown.
6. **Downstream pathway incompletely mapped.** How accumulated AMOTL1 quantitatively perturbs Hippo-YAP vs junction/actin signaling in each affected tissue is not fully dissected.
7. **No therapeutics.** No disease-modifying treatment or trial; the degradation-escape mechanism is a therapeutic hypothesis only.

Proposed Follow-up Experiments / Actions

1. **Natural-history registry** aggregating all reported and new CFCHS individuals to refine phenotype frequencies, penetrance, expressivity, and genotype–phenotype correlations (e.g., R157 vs P160 vs Q161 alleles).

2. **Improved animal models:** characterize zebrafish amotl1 hotspot knock-ins across craniofacial, cardiac, and hepatic lineages; develop conditional/tissue-specific mouse knock-in strategies (e.g., neural-crest-restricted) to overcome the global-mouse discordance.
3. **Single-cell and spatial transcriptomics** of patient-derived iPSC models (neural crest, cardiomyocyte, hepatocyte, endothelial organoids) to map cell-type-specific junction/migration and Hippo-YAP dysregulation.
4. **Quantitative degradation/stability assays** across all hotspot variants to confirm the uniform "degradation-escape" mechanism and test whether degree of stabilization predicts phenotype severity.
5. **Therapeutic proof-of-concept:** test whether restoring mutant AMOTL1 turnover (e.g., degrader/PROTAC approaches, allele-selective knockdown with ASO/siRNA) rescues junction/migration defects in vitro and malformations in zebrafish.
6. **Clinical surveillance guidelines:** formalize a multidisciplinary protocol (cardiac echo, hepatic LFT/imaging, audiology, developmental assessment, craniofacial follow-up) for diagnosed individuals.
7. **ClinVar/panel curation:** ensure all hotspot variants are deposited with functional evidence and ACMG classifications to support rapid clinical diagnosis.

Report compiled from 7 confirmed findings and 10 primary/review papers across 5 investigation iterations. All mechanistic and clinical claims are attributed to the cited PMIDs with verbatim abstract quotes where available.
